

The diagram illustrates the architecture of a fleet management system, organized into three main layers: **infrastructure**, **domain**, and **application**.

infrastructure (Yellow background):

- adapters.inbound.rest**: Contains **dto** (PageResponse, ScheduleRequest, AssignmentRequest; MaintenanceRequest, IncidentRequest, KpiSnapshot), **VehicleController**, **TripController**, **IncidentController**, **MaintenanceController**, **ScheduleController**, **AssignmentController**, **KpiController**, and **DocumentController**.
- adapters.inbound.messaging**: Contains **KernelEventConsumer** and **DocumentExpiryCheckJob**.
- adapters.outbound.persistence**: Contains **entity** (VehicleLocalEntity, ScheduleEntity; AssignmentEntity, KpiSnapshotEntity), **repository** (VehicleLocalR2dbcRepository, ScheduleR2dbcRepository; AssignmentR2dbcRepository, KpiSnapshotR2dbcRepository), **VehiclePersistenceAdapter**, **IncidentPersistenceAdapter**, **KpiPersistenceAdapter**, and **DocumentPersistenceAdapter**. It also includes a note: «uses» «implements».
- adapters.outbound.external**: Contains **RemoteAuthAdapter**, **FakeAuthAdapter**, **GeofenceApiAdapter**, **VehicleApiAdapter**, and **KafkaOperationEventAdapter**.
- config**: Contains **SecurityConfig**, **KafkaConfig**, **RedisConfig**, **ScheduledJobConfig**, and **OpenApiConfig**, **WebClientConfig**.

domain (Light Blue background):

- entity**: Contains **VehicleLocalEntity**, **ScheduleEntity**, **AssignmentEntity**, and **KpiSnapshotEntity**.
- repository**: Contains **VehicleLocalR2dbcRepository**, **ScheduleR2dbcRepository**, **AssignmentR2dbcRepository**, and **KpiSnapshotR2dbcRepository**.

application (Light Green background):

- model**: Contains **Vehicle**, **Driver**, **Fleet**, **IncidentPersistencePort**, **Incident**, and **OperationEventPort**.
- ports**: Contains **Schedule** and **Assignment**.
- exception**: Contains **Schedule** and **Assignment**.
- in**: Contains **ManageVehicleUseCase**, **ManageTripUseCase**, **ManageMaintenanceUseCase**, **ManageIncidentUseCase**, **ManageScheduleUseCase**, and **ManageAssignmentUseCase**.
- out**: Contains **VehiclePersistencePort**, **KpiPersistencePort**, **DocumentPersistencePort**, **ExternalGeofencePort**, **OperationException**, **PlanningException**, **DocumentException**, **KpiException**, **ScheduleService**, **AssignmentService**, **KpiService**, and **DocumentService**.
- service**: Contains **VehicleService**, **TripService**, **MaintenanceService**, **IncidentService**, and **FuelRechargeService**.
- AuthPort**: A separate port package.

Relationships:

- «uses»** relationships: From **VehiclePersistenceAdapter** to **VehiclePersistencePort**, from **IncidentPersistenceAdapter** to **IncidentPersistencePort**, from **KpiPersistenceAdapter** to **KpiPersistencePort**, and from **DocumentPersistenceAdapter** to **DocumentPersistencePort**.
- «implements»** relationship: From **VehicleLocalR2dbcRepository** to **VehiclePersistencePort**.
- «uses»** relationship: From **VehicleLocalEntity** to **VehiclePersistencePort**.
- «uses»** relationship: From **AssignmentEntity** to **AssignmentPersistencePort**.
- «uses»** relationship: From **KpiSnapshotEntity** to **KpiPersistencePort**.
- «uses»** relationship: From **DocumentEntity** to **DocumentPersistencePort**.